

Improving Campus Transport Facilities at FAST NUCES Islamabad

Group Members:

Moiz Ansari 23i-0523

Abdullah Siddique 23i-0617

Shahmeer Amir 23i-0622

Faizan Ur Rehman Khan 23i-3021

Course Name:

Technical & Business Writing

Course Instructor:

Emaan Mirza

Submission Date:

22/04/25

Target Audience:

University Bus Transport Manager

Cover Letter:

Faizan Ur Rehman
23I-3021, CS-F
i233021@isb.nu.edu.pk

April 22, 2025

Muhammad Naveed
Campus Transport Manager
Uzair Transport

Subject: Application for developing a website for FAST Bus Transport Service

Dear Mr. Naveed,

I am most eager to provide you with a proposal for developing an online service for the FAST NUCES Islamabad Bus Transport service. This service will include a multi-dimensional website that will improve the transport experience of the students availing the transport service, as well as the bus drivers, by multiple folds.

The online service will include an online tracking system for both the buses on various routes, as well as the transporters, in order to facilitate both parties in making the entire experience more seamless. It will also include online transaction services that will rid the students as well as the company from having to deal with in-person transactions or any need for physical proof of payment.

I humbly request you to go through with this proposal and make a decision that will be sustainable and improve your business model ten fold, as well as improve the user experience of thousands of students and dozens of drivers. Looking forward to your response.

Sincerely,
Faizan Ur Rehman Khan

Cover Letter:	2
List of Illustrations:	3
Abstract:	3
INTRODUCTION	3
Overview:	3
Background:	4
Problems:	4
Needs:	4
Limitations:	5
Body:	6
Methodology:	6
Technologies:	6
Features:	7
Development Methodology:	7
Cost:	8
Personnel:	9
Feasibility:	9
Conclusion:	10
Works Cited:	10

List of Illustrations:

Abstract:

This proposal is for introducing an online service within Uzair Transport in order to facilitate the students as well the drivers within the company. This includes benefits like real-time bus tracking, polling for democratic bus allocation, driver contact details as well as personalized booking systems. The details regarding the various technologies that will be used by the website as well as the feasibility of the various resources used like HTML, JS, and Node.JS have been detailed for an informed decision.

INTRODUCTION

Overview:

In the contemporary digital age, technology has changed how people access and spread information. The newer generation, in particular, are much more reliant on technology for accessing information. As a result, most services in the university have adopted an online

presence, One-Stop service having their dedicated email addresses, or the career counselors having their own webpages. With the increase in popularity of the transport department, Uzair Transport, it is high time that they too explore a digital option. A well-designed, user-friendly website can significantly enhance the efficiency of the transportation system and improve the daily commuting experience for students.

Background:

Currently, the management system of the transport department is inefficient and unprofessional. A simple Whatsapp channel is used to convey departure times and emergency notices. While this may be convenient for the department, it is unable to meet the needs of students. Important messages that are uploaded to the channel are lost in the sea of irrelevant texts. The departure times are always subject to change causing confusion among students about the bus timings.

While there are general-purpose apps available, like Whatsapp, they usually lack customization, flexibility, and transport-specific features required for effective management of campus transportation services. A customizable website would directly address these gaps and offer specific functionalities that a WhatsApp group or any other third-party application cannot provide.

Problems:

There are several problems that the transport department have which could be solved by a website.

- **Limited online presence** has made accessing bus schedules, assigned buses and updates challenging
- **Inefficient communication** between the transport department and students have led to a lot of mismanagement. 92% of the 10 students utilizing the university transport have reported either missing their buses or having to cram into overcrowded buses due to miscommunication.
- Due to the **lack of real-time updates**, students are forced to stand outside in severe weather conditions waiting for their bus allocations. This problem will become much more severe in the coming days as summer approaches.

With a proper website, these issues can be easily resolved.

Needs:

A dedicated website will offer a centralized system for the transport department to share information like diver details, bus timings and allocations and for students to access these

information in a user friendly manner. Furthermore, a customizable website would ensure efficient bus allocation which overcomes the problem of overcrowded buses. Additionally, due to the personalized nature of websites, the user experience will be far better on websites as compared to a whatsapp channel. Similarly, the transport department will benefit from this as well. Firstly, managing students, updates and routes will be much easier due to the availability of structured website content. Secondly, a website will enhance the image of the university transport services, by presenting a more modern and professional front to students, faculty and visitors.

Scope:

The proposed transport management website will emphasize resolving the existing inefficiencies and streamlining the communication system between students and the transport department. The major features of the platform will be:

- **Personal User accounts** for students, where they can view their assigned bus and departure time.
- A **polling system** where students can vote for their preferred departure times, allowing the department to plan more during bus allocation. This solves the biggest problem i.e overcrowded buses.
- **Live tracking of buses** using location services, providing students with real-time updates on bus arrivals. This is really helpful for students whose parents pick and drop them at the bus stations.
- **Driver profiles and contact information**, this allows students to directly contact their drivers in case of specific instructions.
- **Instant notifications and announcements** for schedule changes or emergencies.
- An **organized dashboard** for transport department staff to monitor bus allocation, adjust schedules, and communicate with students.

Limitations:

- The initial release will concentrate on fundamental scheduling, polling, and alerting, with features such as live tracking and route optimization added in subsequent releases.
- The user accounts will be restricted to university domain emails to avoid cyberattacks and enhance data security.

Body:

Methodology:

Technologies:

For the website solution, four primary technologies will be used. It includes HTML (Hyper Text Markup Language), CSS (Cascading Style Sheet), Javascript and Node.js. Their purpose is stated as follows:

HTML: This is the first component of the front-end. It will be used for structuring and adding relevant details about the website. It will basically answer “What is this website about?”

1. Adding relevant information on the website like website name and headline and adding images.
2. Adding buttons and forms for data and students’ requests entry.
3. Structuring the information into meaningful blocks on a page e.g, a block containing buses names and pictures while another block contains the buses schedules.

CSS: The second component of the front-end which will be used for styling the website. It will be utilized for adding transitions and transformations for improving the UI/UX (User Interface/User Experience).

1. Adding styles to text and images added in the HTML which includes changing colors and sizes.
2. Setting background colors or images to make the website look pleasant to use.
3. Styling forms and buttons to gain students attention.

Javascript: A very crucial component of the front-end that will provide functionality to the website. The functionality will help provide our website some meaning. It will help students navigate through the schedules, view available buses, book seats, and select their location.

1. Changing the information, elements or images on run-time e.g, a student navigating through a list of buses. Whenever the student clicks on the button next, javascript will change the elements and display the next information.
2. Updating the status (available, left, out of order, etc) of the buses.

Node.js: This technology will be used at the backend which will provide the data in real-time. Node.js will help in following things:

1. Connect the database of buses, drivers and locations with the front-end.
2. Retrieve and provide required data to the students in the real-time.
3. Handling multiple students at the same time.
4. Avoiding conflicts between concurrent requests.
5. Updating the database.

Features:

The website will include the following features:

Login System: Students will login with their NU email address and a specific password assigned to them by the bus administrator. This password allocation will allow only those students who use the university transport to use the website.

Bus Schedules: The arrival and departure timings of buses will be listed which will help students prepare themselves accordingly.

Booking: Students will be able to book time slots beforehand indicating that they will travel on that specific day and time.

Email Based Notifications: Students who booked their time slots will be notified through an email as a reminder which will help reduce their chances of missing the bus.

Buses Arrangements: The number of students who booked their slots will help drivers to decide the number of buses to make operational beforehand.

Bus Tracking on Maps: Students waiting on the stops will be able to track the location of their bus.

Drivers Information: Drivers phone numbers will be available for students to contact in case of emergency.

Development Methodology:

The development methodology that will be opted is Agile methodology. Since most of the components of this project are linked with each other, it is necessary to perform activities in an iterative manner. This will help in dealing with the challenges of one module (like HTML) before progressing to the next one (like CSS) so that the previous work is completely available for the next ones to build on.

Following is the workflow of Agile methodology:

1. **Planning:** Gathering the requirements and features necessary for the project. It includes student registration, bus schedules feature, etc.
2. **Sprint-Phase:** Prioritizing the essential features that have to be implemented first and divide it among the team. For instance, prioritizing student registration and dividing this task among the team members.

3. **Development-Phase:** Developing the assigned tasks and then conducting meetings on a daily basis to keep track of the progress. Eg, after completing the registration task, team members discuss the shortcomings of the registration task in meetings.
4. **Review:** The process is reviewed for future improvements and this cycle is repeated.

Cost:

To ensure that the website for “**Improving Campus Transport**” initiative is both functional and sustainable, we have assessed the following costs:

- **Domain & Hosting:**

To minimize the costs and attain student-friendly offers, we plan to register a free “.site” domain which is included in a bundle package. The package not only provides a hosting service with free domain but also provides free domain privacy for protection from potential spam and identity theft. Apart from that it offers free SSL (Secure Sockets Layer) certificates for the first year, a free personal email account as well as a free website builder. All of this is available at just **\$22.88/yr** for the first year (Namecheap, 2025). Afterwards it will be increased to **\$48.88/yr** which is incredibly low as compared to the individual pricing plan, which would be approximately \$150/yr.

- **Development & Maintenance Costs:**

As there are four members in the group, and everyone has expertise in web development, we plan to develop and maintain the website by ourselves through HTML, CSS and JavaScript. Each member will then have a responsibility to maintain specified parts of the website afterwards. So, the development and Maintenance cost would be **\$0**.

- **Website Designer:**

Initially, we had planned to hire a freelance UI/UX designer to design a user-friendly and accessible interface. But since we are availing the all-in-one package for hosting and domain, we have access to a free website designer with **200+ designs** to start with. Hence, there would be no additional cost for website designing.

- **Any additional expenses:**

- **Bus Trackers**

To have access to live locations of buses, we plan to hire online GPS trackers, which will cost **\$53.46/yr** (eTracking.pk, 2025). This will allow everyone to effectively monitor campus shuttles without facing significant expenses.

- o **Booking Forms**

Instead of relying on third-party apps, we will develop a custom booking system with our coding skills in JavaScript. This will allow us to reduce the implementation costs and have full control over its functionality and design.

- **Total Estimated Cost:**

- o **Initial Year:** \$22.88 (hosting) + \$53.46 (GPS) = **\$76.34/year**

- o **Subsequent Years:** \$48.88 (renewal) + \$53.46 (GPS) = **\$102.34/year**

- **Works Cited:**

1. **Namecheap – Domain & Hosting**

Namecheap. (2025). Domain and Hosting Bundle.¹

2. **eTracking.pk – GPS Tracker Pricing**

eTracking.pk. (2025). HTV Truck Tracker Pricing.²

Personnel:

Front-end: Moiz and Abdullah will be working with the front-end. Moiz with in-depth knowledge of UI will be handling HTML and CSS while Abdullah will be working with the javascript to make the website functional and improving UX. Both the team members will collaborate with each other for a better performance.

Backend: Shameer and Faizan will be dealing with the backend functionality of the website. They will collaboratively work on Node.js to handle databases and its interaction with the front end.

Feasibility:

Survey showed that around **67%** of students who use the transport facility of the university are not satisfied with it due to disoriented management. This situation gets worse in summers, examination days and specially on events like NASCON and Competitions nights.

Currently, the number of buses are decided by keeping in view three timings of departure. Fixed number of buses for each timing causes a major problem. The problem is overcrowding in the buses at one time while the other timings, buses may not have enough students. The survey showed that **67%** of the students face the overcrowding issue almost an entire week.

Further, the students find it difficult to navigate their buses as they have to find the drivers and ask for their bus in the scorching heat of summers and chilling winds of winters. This problem

¹ Retrieved April 21, 2025, from

<https://www.namecheap.com/hosting/domain-and-hosting-bundle/>

² Retrieved April 21, 2025, from <https://etracking.pk/htv-truck-tracker-pricing/>

will be solved as the students will be notified about their respective bus before the departure time.

Our survey found that **67%** of the students will find such website useful in their daily commute to university.

Advantages:

1. Through our web solution, the bus division will be entirely based upon the number of students available at a specific time, eliminating the overcrowding issue.
2. Survey showed that 67% of students will benefit from the pre-booking functionality of our solution and are ready with the rules and regulations that come with it, like a fine system for students who do not show up on a booked time slot.
3. The university's transport management will eventually benefit from the website as it will help the drivers make decisions beforehand improving the students' experience.
4. A Student's good experience will increase the university's transport reputation, bringing it more students in future.
5. Ensures smooth execution of the daily commute events with no hindrance at both the student and drivers end.

Conclusion:

The online service for Uzair Transport will allow the students and the drivers to have real-time control over what mode of transport they want to avail and when. This will solve the long-standing problem of the lack of real-time updates as well as the dependence on in-person updates without a centralized source that can ensure both parties receive updates remotely. If executed properly, this project can be a game changer for both parties involved as well as Uzair Transport and their business model, making it more sustainable with a higher quality of service.

Works Cited:

1. <https://www.namecheap.com/hosting/domain-and-hosting-bundle/>
2. <https://etracking.pk/htv-truck-tracker-pricing/>